

1. Overview

As part of my independent development in data analytics, I have explored the fundamentals of Microsoft Power BI and developed an introductory understanding of how business intelligence tools can be used to transform data into useful reports and visualisations.

My learning has focused on understanding the purpose of Power BI, the structure of reports and dashboards, appropriate data visualisation, interactive filtering and introductory Data Analysis Expressions (DAX).

This document records my self-directed learning and demonstrates my foundational understanding of these concepts. It does not represent advanced or professional Power BI development experience.

2. Understanding the Power BI Workflow

My understanding of a basic Power BI workflow can be summarised as:

Data Source → Data Preparation → Data Model → DAX Measures → Visualisations → Report

A dataset can first be imported from a source such as an Excel workbook or CSV file.

The data may then require cleaning and transformation before analysis. Fields can be organised into a data model and calculations can be created using DAX.

The resulting information can then be displayed using visualisations such as charts, cards and tables. Filters and slicers can allow users to interact with the report and investigate particular categories or periods.

3. Reports, Dashboards and Visualisations

I developed an understanding of the role of different elements within Power BI.

Report: A report presents data through one or more pages containing interactive visualisations. Reports allow users to explore information and investigate different aspects of a dataset.

Dashboard: A dashboard provides a high-level visual overview of important information and key performance indicators.

Visualisation: A visual representation of data, such as a bar chart, line chart, card or table. Different visualisations are appropriate for different analytical questions.

For example, a bar chart can be useful when comparing values between companies or sectors, while a line chart is more appropriate for examining a trend over time.

4. Sample Financial Report Design

To apply these concepts, I designed a conceptual financial performance report based on a sample dataset containing information such as:

- Company
- Sector
- Revenue
- Cost
- Profit
- Profit Margin
- Date

The report could contain the following visualisations.

Total Revenue Card

A card could display total revenue as a key performance indicator, allowing the user to immediately see the overall revenue represented within the current report context.

Total Profit Card

A second card could display total profit, providing a high-level indication of overall financial performance.

Revenue by Company – Bar Chart

A bar chart could compare revenue between companies. This would make differences in company performance easier to identify visually.

Revenue by Month – Line Chart

A line chart could display revenue over time. This would help identify increases, decreases and other trends across different periods.

Profit by Sector – Bar Chart

A bar chart could compare total profit across sectors such as Technology, Banking and Automotive.

Sector Slicer

A sector slicer could allow a user to select a particular sector and focus the report on the relevant information.

5. Filters and Slicers

Filters allow users to restrict the information displayed in a report according to selected criteria.

A slicer provides an interactive way of applying filtering directly from the report.

For example, if a financial report contained companies from the Technology, Banking and Automotive sectors, a **Sectors** slicer could allow the user to select **Technology**.

The report's relevant visualisations could then respond to the filter context and display information relating to the selected sector.

This would allow users to investigate specific parts of the dataset without having to create a separate report for every category.

6. Introduction to DAX

DAX stands for **Data Analysis Expressions** and is used in Power BI for calculations involving data within a model.

As part of my introductory learning, I explored how simple measures could be used to summarise financial information.

For a hypothetical table named FinancialData, an example measure for total revenue would be:

Total Revenue =
SUM(FinancialData[Revenue])

The SUM function adds the values contained in the Revenue column and returns total revenue within the current filter context.

A similar measure could calculate total cost:

Total Cost =
SUM(FinancialData[Cost])

This aggregates the values in the Cost column.

7. Calculating Profit Using Measures

Measures can also reference other measures.

For example:

Total Profit =
[Total Revenue] - [Total Cost]

This calculates total profit by subtracting the Total Cost measure from the Total Revenue measure.

Using separate measures makes the calculation easier to understand and allows the measures to be reused within other calculations and visualisations.

8. Calculating Profit Margin

An introductory profit-margin measure could be written as:

Profit Margin % =
DIVIDE([Total Profit], [Total Revenue], 0)

The measure divides total profit by total revenue.

Using DIVIDE also allows an alternative result to be specified when division cannot be performed because the denominator is zero. In this example, the alternative result is 0.

The resulting measure could then be formatted as a percentage within Power BI.

For example, if total revenue was £120,000 and total profit was £40,000:

Profit Margin = £40,000 ÷ £120,000 = 33.33%

This measure could be displayed using a card or analysed across different companies or sectors.

9. Understanding Filter Context

An important concept I encountered while learning introductory DAX was **filter context**.

A measure does not necessarily produce the same result in every part of a report because the data being evaluated can be affected by filters.

For example, the measure:

Total Revenue =
SUM(FinancialData[Revenue])

could calculate revenue across the complete dataset when no sector is selected.

If a user selected **Banking** using a slicer, the same measure could instead return revenue within the Banking filter context.

This helped me understand the relationship between DAX measures and interactive report filtering.

10. Identifying Key Trends

A well-designed business intelligence report should make important information easier to interpret rather than simply displaying large quantities of raw data.

Using the proposed financial report, I could investigate questions such as:

- Which company generates the highest revenue?
- Which sector generates the most profit?
- How does revenue change over time?
- What is the overall profit margin?
- How do results change when an individual sector is selected?

Cards could summarise key performance indicators, while charts could reveal comparisons and trends.

Filters and slicers could then allow the user to investigate particular areas of interest.

11. Relationship to My Existing Data Analytics Skills

My introductory Power BI learning complements my development in other data analytics technologies.

I have worked with **Excel** for financial analysis and data summarisation and developed skills in **Python/Pandas and SQL** for working with structured datasets.

Power BI introduces another aspect of the analytics process by focusing on presenting information through interactive business intelligence reports and visualisations.

Understanding introductory DAX also develops my understanding of how calculated measures can be used to summarise data dynamically.

12. Current Skill Level and Further Development

My current Power BI knowledge is **foundational and based on self-directed learning**.

I understand the basic purpose of Power BI reports and dashboards, the use of common visualisations, the role of filters and slicers, and introductory DAX concepts including measures, aggregation and filter context.

My next stage of development would be to apply these concepts directly within Power BI by importing a dataset, creating a data model, implementing DAX measures and constructing an interactive report when I have suitable access to the Power BI environment.

13. Conclusion

Through independent learning, I have developed a foundational understanding of Power BI and its role within data analytics and business intelligence.

I understand how different visualisations can be selected to communicate information, how filters and slicers can support interactive analysis, and how introductory DAX measures can calculate metrics such as revenue, cost, profit and profit margin.

This independent learning complements my existing development in Excel, Python, SQL and financial data analysis and provides a foundation for further practical development using Microsoft Power BI.